

UNIT-IV

Development of surfaces

A development is the unfold / unrolled flat / plane figure of a 3-D object. It is also called a pattern where the plane may show the true size of each area of the object. When the pattern is cut, it can be rolled or folded back into the original object as shown in figure 1.

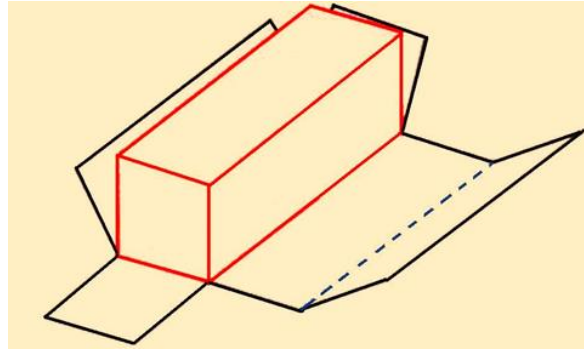


Figure 1. Typical development of the surface of a cuboid.

Types of development

There are three major types of development followed by industries. Examples are shown in figure 2.

1. Parallel line development: In this parallel lines are used to construct the expanded pattern of each three-dimensional shape. The method divides the surface into a series of parallel lines to determine the shape of a pattern.
2. Radial line development: In this, lines radiating from a central point to construct the expanded pattern of each three-dimensional shape is used. These shapes each form part of a cone and lines radiating from the vertex of the cone generate the expanded pattern of the curved surface as shown in the following explorations.
3. Triangulation method: This is generally used for polyhedron, single curved surfaces, and warped surfaces.
4. Approximate development: In this, the shapes obtained are only approximate. After joining, the part is stretched or distorted to obtain the final shape

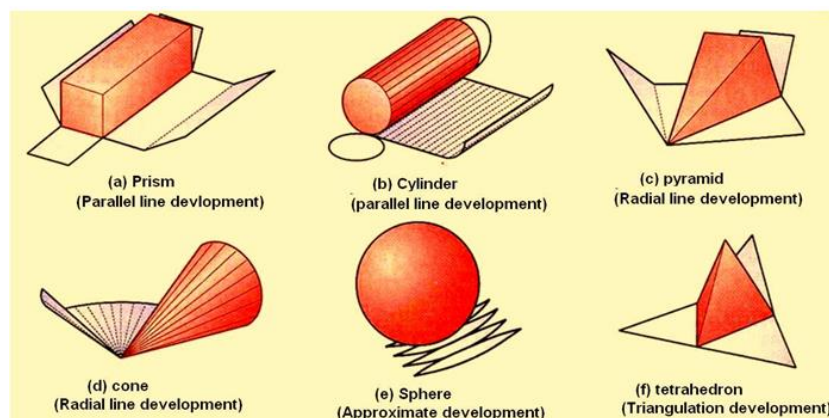


Figure 2. Typical examples of the various types of development.

A true development is one in which no stretching or distortion of the surfaces occurs and every surface of the development is the same size and shape as the corresponding surface on the 3-D object. e.g. polyhedrons and single curved surfaces.

As illustrated in figure 3, polyhedrons are composed entirely of plane surfaces that can be flattened true size onto a plane in a connected sequence, whereas single curved surfaces are composed of consecutive pairs of straight-line elements in the same plane which is obtained for a cone.

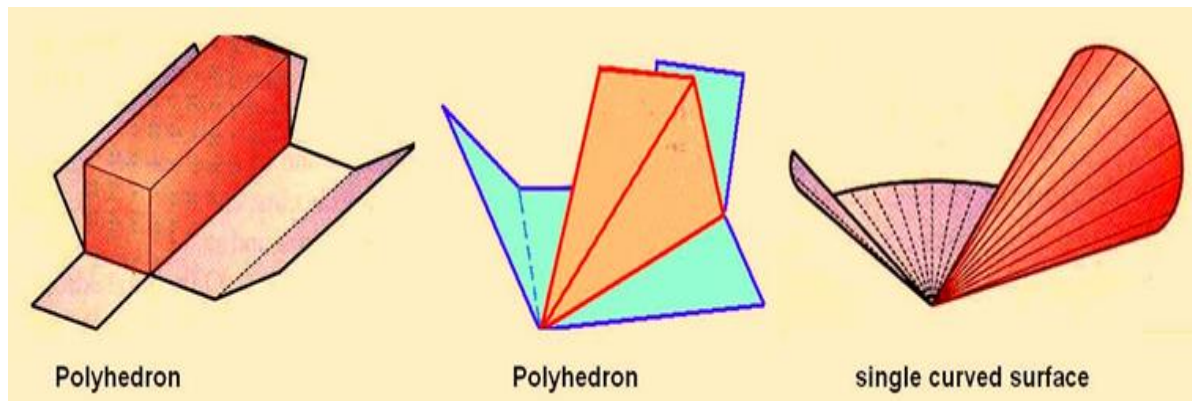


Figure 3. Shows the true development obtained for polyhedrons and single curved surface

An approximate development is one in which stretching or distortion occurs in the process of creating the development. The resulting flat surfaces are not the same size and shape as the corresponding surfaces on the 3-D object. Wrapped surfaces do not produce true developments, because pairs of consecutive straight-line elements do not form a plane. Also double-curved surfaces, such as a sphere do not produce true developments, because they do not contain any straight lines. An example of the approximate development of a sphere is shown in figure 4 .

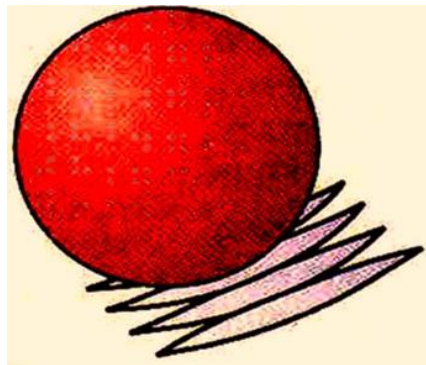


Figure 4 showing an approximate development of a sphere.

Classifications of developments

1. Parallel-line development:

They are made from common solids that are composed of parallel lateral edges or elements. e.g. Prisms and cylinders as shown in figure 5. The cylinder is positioned such that one element lies on the development plane. The cylinder is then unrolled until it is flat on the development plane. The base and top of the cylinder are circles, with a circumference equal to the length of the development. All elements of the cylinder are parallel and are perpendicular to the base and the top. When cylinders are developed, all elements are parallel and any perpendicular section appears as a stretch-out line that is perpendicular to the elements.

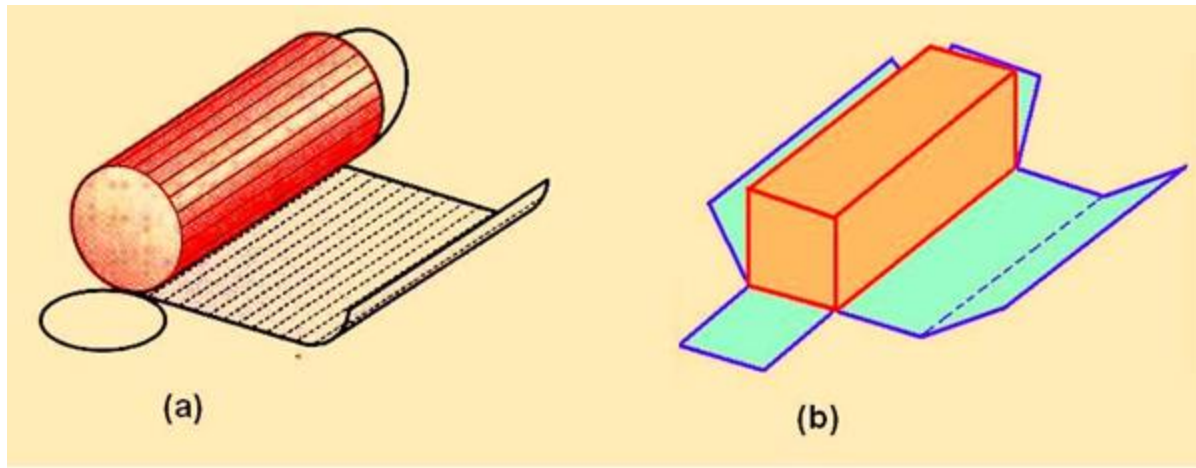


Figure 5 shows the parallel line development technique for (a) cylinder and (b) rectangular block.

2. Radial-line development

Radial-line developments are made from figures such as cones and pyramids. In the development, all the elements of the figure become radial lines that have the vertex as their origin. Figure 6 shows the radial development for a cone. The cone is positioned such that one element lies on the development plane. The cone is then unrolled until it is flat on the development plane. One end of all the elements is at the vertex of the cone. The other ends describe a curved line. The base of the cone is a circle, with a circumference equal to the length of the curved line.

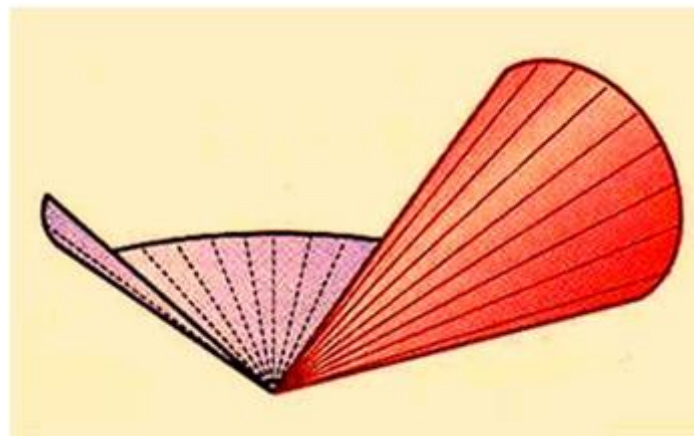


Figure 6 shows the radial development method for a cone.

3. Triangulation developments:

Made from polyhedrons, single-curved surfaces, and wrapped surfaces. The development involve subdividing any ruled surface into a series of triangular areas. If each side of every triangle is true length, any number of triangles can be connected into a flat plane to form a development. This is illustrated in figure 7 for a triangular pyramid. Triangulation for single curved surfaces increases in accuracy through the use of smaller and more numerous triangles. Triangulation developments of wrapped surfaces produces only approximate of those surfaces.

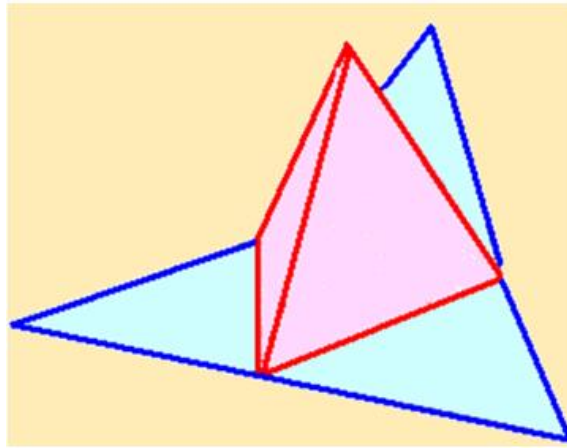


Figure 7 shows the triangulation method for obtaining the development of a triangular pyramid.

4. Approximate developments

Approximate developments are used for double curved surfaces, such as spheres. Approximate developments are constructed through the use of conical sections of the object. The material of the object is then stretched through various machine applications to produce the development of the object. This is illustrated in figure 4.

Parallel-line developments:

Developments of objects with parallel elements or parallel lateral edges begins by constructing a stretch-out line that is parallel to a right section of the object and is therefore, perpendicular to the elements or lateral edges. Figure 8 illustrates the steps followed for obtaining the development of a rectangular prism by parallel line development. In the front view, all lateral edges of the prism appear parallel to each other and are true length. The lateral edges are also true length in the development. The length, or the stretch-out, of the development is equal to the true distance around a right section of the object.

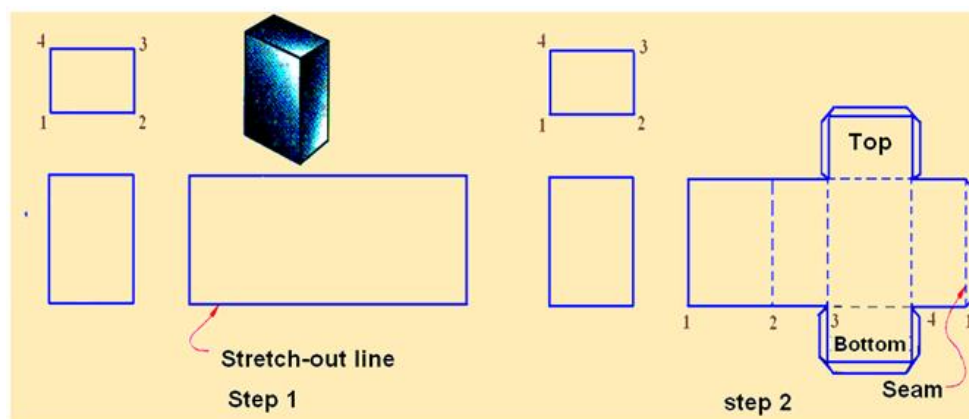


Figure 8. Stepwise procedure for obtaining the development of a rectangular prism.

Step 1. To start the development, draw the stretch-out line in the front view, along the base of the prism and equal in length to the perimeter of the prism. Draw another line in the front view along the top of the prism and equal in length to the stretch-out line. Draw vertical lines between the ends of the two lines, to create the rectangular pattern of the prism.

Step 2. Locate the fold line on the pattern by transferring distances along the stretch-out line in length to the sides of the prism, 1-2, 2-3, 3-4, 4-1. Draw thin, dashed vertical lines from points 2, 3, and 4 to represent the fold lines. Add the bottom and top surfaces of the prism to the development, taking measurements from the top view. Add the seam to one end of the development and the bottom and top.

1. Development of a truncated prism

The method of obtaining the development of a truncated prism is shown in figure 1.

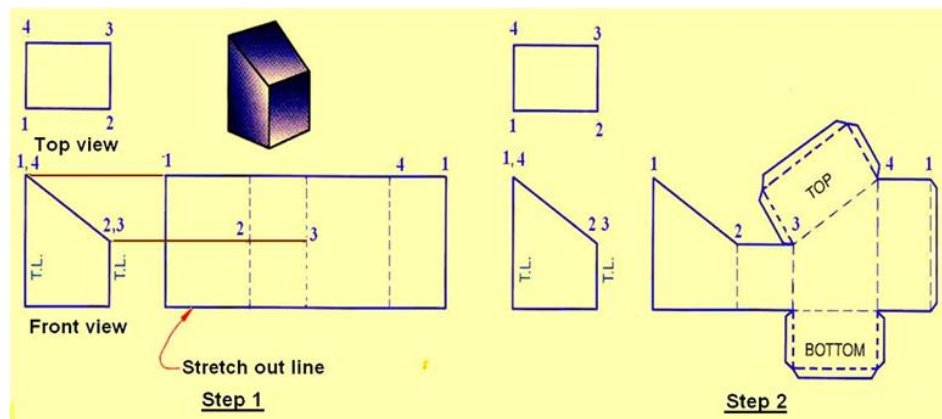


Figure1. Shows the development of a truncated rectangular prism.

Step 1: Draw the stretch-out line in the front view, along the base of the prism and equal in length to the perimeter of the prism.

Locate the fold lines on the pattern along the stretch-out line equal in length to the sides of the prism, 1-2, 2-3, 3-4, and 4-1.

Draw perpendicular construction lines at each of these points.

Project the points 1, 2, 3, and 4 from the front view

Step 2: Darken lines 1-2-3 and 4-1. Construct the bottom and top, as shown and add the seam to one end of the development and the top and bottom

2. Development of a right circular cylinder

The Procedure for obtaining the development of a cylinder is illustrated in figure 2.

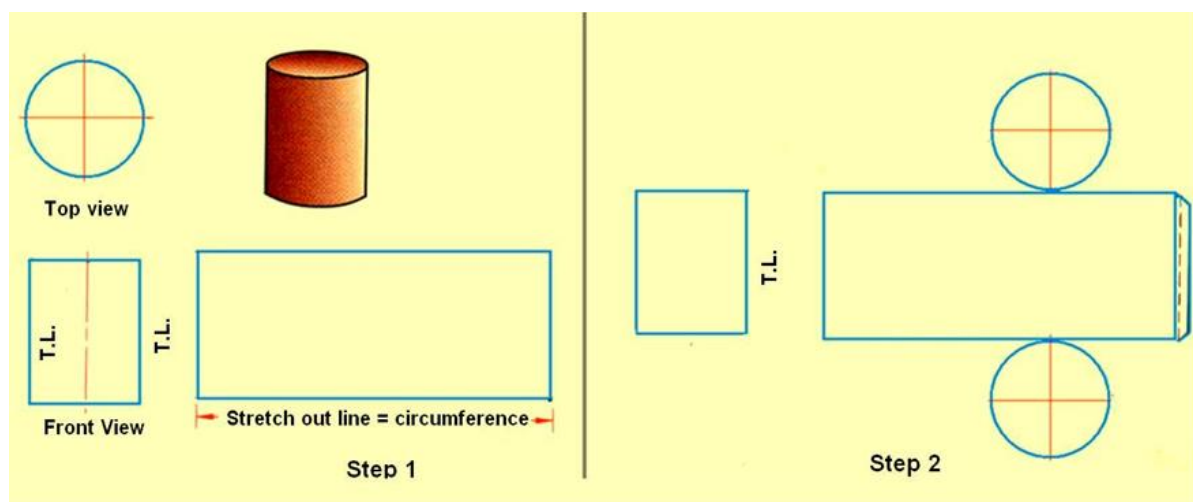


Figure 2. Shows the step wise procedure for obtaining the development of a cylinder.

Step 1. In the front view, draw the stretch-out line aligned with the base of the cylinder and equal in length to the circumference of the base circle.

At each end of this line, construct vertical lines equal in length to the height of the cylinder.
Step 2. Add the seam to the right end of the development, and add the bottom and top circles.

3. Development of a truncated right circular cylinder

The development of a truncated cylinder is shown in figure 3.

Step1. The top view and front view of the cylinder is drawn. The stretch out line, is aligned with the base in the F.V., is drawn with length equal to the circumference of the cylinder. Construct the rectangle with the stretch out line as one length and height of the cylinder as the width.

Step2. The top circular view of the cylinder then divided into a number of equal parts . The stretch-out line is also divided into 12 equal parts from which vertical lines 1, 2, 3, 4, 12 are constructed. The intersection points in the T.V. are projected into the F.V. . Draw horizontal projectors from points 1, 2, 3,, 12 to intersect the vertical lines 1,2, 3, 4, 12 in the stretch out line. , where the projected lines intersect the angled edge view of the truncated surface of the cylinder. The intersections between these projections and the vertical lines constructed from the stretch-out line are points along the curve representing the top line of the truncated cylinder. Join the intersection points with a smooth curve to obtain the development of the lateral surface of the cylinder.

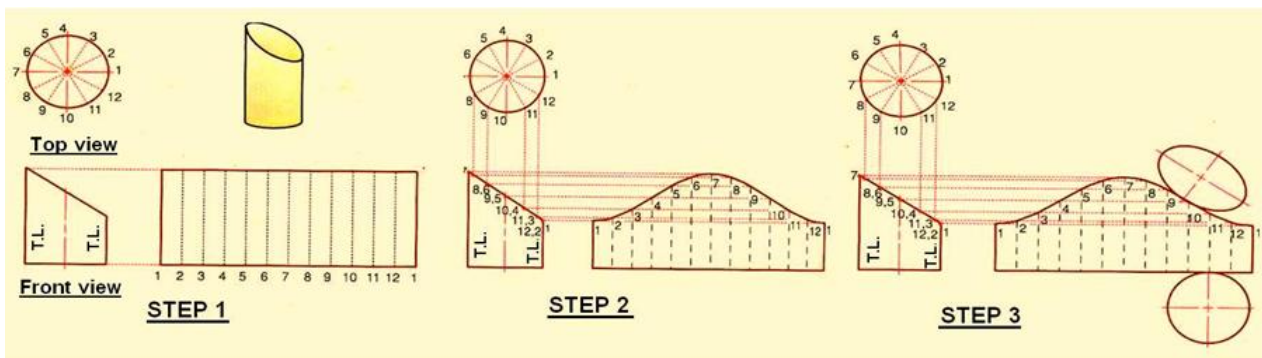


Figure 3. Shows the development of a truncated cylinder.

Step 3. Draw the circle with diameter equal to the diameter of the cylinder at any point on the base of the development to obtain the development of the base surface of the cylinder. Draw an ellipse with the truncated length (length 1-7 in the step 2) as major diameter and diameter of the cylinder as the minor diameter on the top part of the development to obtain the final development of the surfaces of the truncated cylinder.

4. Development of a right circular cone

The development of a cone is shown in figure 4. For a cone, the front view will be a triangle with the slant edge showing the true length of the generator of the cone. To begin this development, use a true-length element of the cone as the radius for an arc and as one side of the development. Draw an arc whose length is equal to the circumference of the base of the cone. This can also be determined by angle $\theta = (r/l) * 360^\circ$, where, r is the radius of the base of the cone and l is the true length of the slant edge. Draw another line from the end of the arc to the apex and draw the circular base to complete the development.

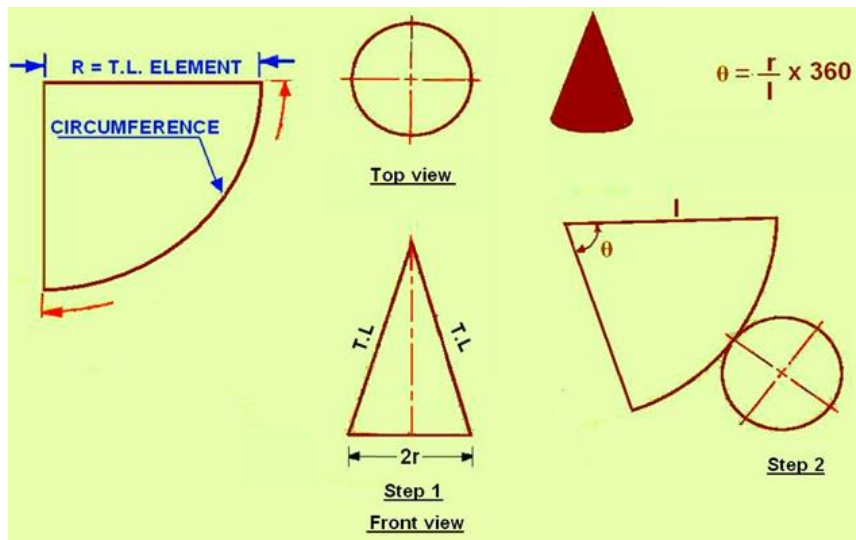


Figure 4. Development of a right cone.

5. Development of a truncated cone:

A cone of base diameter 40 mm and slant height 60 mm is kept on the ground on its base. An AIP inclined at 45° to the HP cuts the cone through the midpoint of the axis. Draw the development.

Solution:

The development of the truncated cone is shown in figure 5.

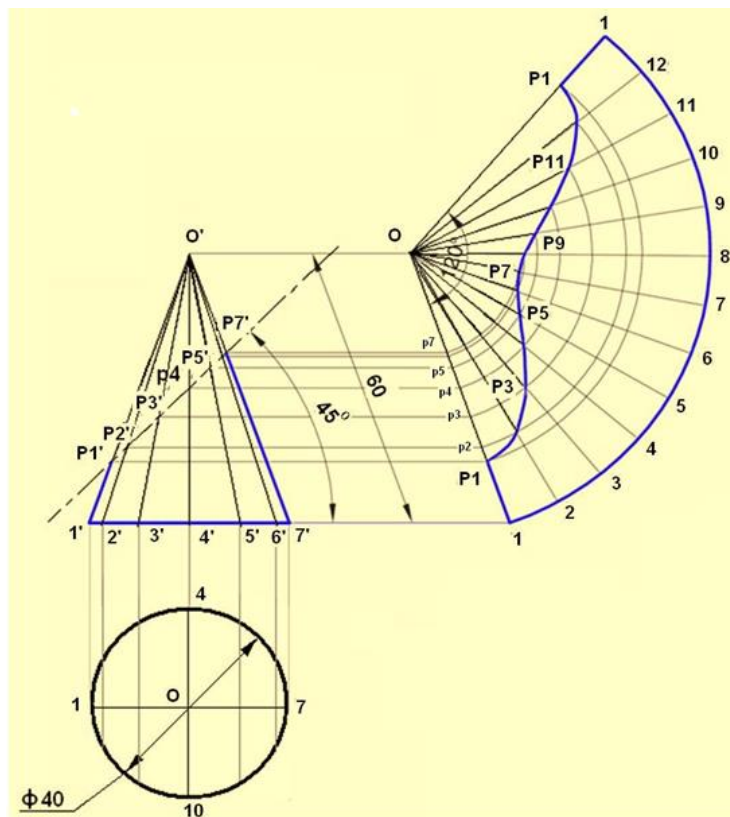


Figure 5. Development of the truncated cone (problem 5).

- Draw the Front view and top view of the cone.
- Divide the circumference of the circle (Top View) into 12 equal parts 1, 2, 3, 4, ..., 12.

- Project these points on the Front view to obtain the points 1', 2', 3', ..., 12'.
- Draw a line inclined at 45 ° to the horizontal and passing through the mid point of the axis of the cone to represent the AIP.
- The locate the intersection points of the AIP with the generators O'-1', O'-2', O'-12' as P1', p2', p3', P12'.
- Draw the projection (figure shown on the right of the Front view) by drawing the line O1 parallel to O' 7'.
- Obtain the included angle of the sector. $\theta = (20/60) * 360 = 120^\circ$ (following the procedure shown in problem 4).
- Then draw sector O-1-1- O with O as a centre and included angle 120°. Divide the sector into 12 equal parts (i.e., 10° each). Draw lines O-2, O-3, O-4, ..., O-12.
- Draw horizontal projectors from P1', P2', ..., P12' such that it meets the line O1 at p1, p2, p3, ..., p12.
- With O as centre and radius O'P1', mark point P1 on line O1.
- With O as centre and radius equal to Op2, draw an arc to intersect the radial line O2 at point P2.
- Similarly obtain points P3, P4, ..., P12, and P1. Join points P1, P2, P3, ..., P12 and P1 to obtain the development of the truncated cone.

Intersection of solids

Whenever two or more solids combine, a definite curve is seen at their intersection. This curve is called the curve of intersection (COI). Lines of intersection are a common feature in engineering applications or products. Figure 1 shows few examples of intersection lines frequently observed in chemical plants, domestic appliances, pipe joints, etc. Curves of intersections are important from the point of view of production of components for engineering applications.



Figure 1. Photographs of engineering components showing curves of intersection.

Cases of Intersection

The cases of intersection depend on the type of intersecting solids and the manner in which they intersect. Two intersecting solids may be of the same type (e.g., prism and prism) or of different types (e.g., prism and pyramid). The possible combinations are shown in table 1.

Table 1. Different cases of intersection

		Solid – 1				
		Prism	Pyramid	Cylinder	Cone	Sphere
Solid – 2	Prism	Case 1	---	---	---	---
	Pyramid	Case 2	Case 6	---	---	---
	Cylinder	Case 3	Case 7	Case 10	---	---
	Cone	Case 4	Case 8	Case 11	Case 13	---
	Sphere	Case 5	Case 9	Case 12	Case 14	Case 15

The two solids may intersect in different ways. The axes of the solids may be parallel, inclined or perpendicular to each other. The axes may be intersecting, offset or coinciding. Therefore, the following sub-cases exist:

- (i) Axes perpendicular and intersecting
- (ii) Axes perpendicular and offset
- (iii) Axes inclined and intersecting
- (iv) Axes inclined and offset
- (v) Axes parallel and coinciding
- (vi) Axes parallel and offset

The type of intersection created depends on the types of geometric forms, which can be two- or three-dimensional. Intersections must be represented on Multiview drawings correctly and clearly. For example, when a conical and a cylindrical shape intersect, the type of intersection that occurs depends on their sizes and on the angle of intersection relative to their axes as well as relative position of their axes. The line of intersection is determined using auxiliary views and cutting planes.

Methods – (1) Line and (2) Cutting-plane methods

Line method: A number of lines are drawn on the lateral surface of one of the solids and in the region of the line of intersection. Points of intersection of these lines with the surface of the other solid are then located. These points will lie on the required line of intersection. They are more easily located from the view in which the lateral surface of the second solid appears edgewise (i.e. as a line). The curve drawn through these points will be the line of intersection.

Cutting-plane method: The two solids are assumed to be cut by a series of cutting planes. The cutting planes may be vertical (i.e. perpendicular to the H.P.), edgewise (i.e. perpendicular to the V.P.) or oblique. The cutting planes are so selected as to cut the surface of one of the solids in straight lines and that of the other in straight lines or circles.

Intersection of two prisms

Prisms have plane surfaces as their faces. The line of intersection between two plane surfaces is obtained by locating the positions of points at which the edges of one surface intersect the other surface and then joining

the points by a straight line. These points are called vertices. The line of intersection between two prisms is therefore a closed figure composed of a number of such lines meeting at the vertices. The method of obtaining intersection lines are shown by means of problem.

Problem 1. A vertical square prism, base 50 mm side, is completely penetrated by a horizontal square prism, base 35 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the prism, while the faces of the two prisms are equally inclined to the prism. Draw the projections of the solids, showing lines of intersection. (Assume suitable lengths for the prisms.)

Solution

The intersection obtained is shown in figure 2. Draw the projections of the prisms in the required position. The faces of the vertical prism are seen as lines in the top view. The points of intersection in this view can be located by the following method.

- Lines 1-1 and 3-3 intersect the edge of the vertical prism at points p_1 and p_3 (coinciding with a).
- Lines 2-2 and 4-4 intersect the faces at p_2 and p_4 respectively.
- The exact positions of these points along the length of the prism may now be determined by projecting them on corresponding lines in the front view.
- For example, p_2 is projected to p_2' on the line $2'2'$. Note that p_4' coincides with p_2' .
- Similarly other points are obtained. Draw lines $p_1'p_2'$ and $p_2'p_3'$.
- Lines $p_1'p_4'$ and $p_3'p_4'$ coincide with the front lines.
- These lines show the line of intersection.
- Lines $q_1'q_2'$ and $q_2'q_3'$ on the other side are obtained in the same manner.
- Note that the lines for the hidden portion of the edges are shown as dashed lines.
- The green lines in the figure represents construction lines.
- The portions $p_1'p_3'$ and $q_1'q_3'$ of vertical edges $a'a'$ and $c'c'$ do not exist and hence, must be removed or kept fainter.

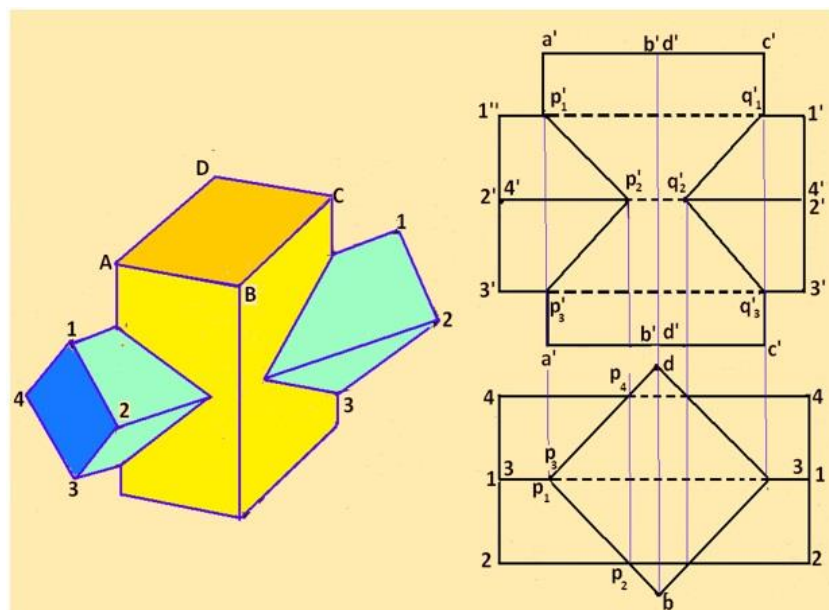


Figure 2 The lines of intersection obtained for problem 1.

Intersection of Cylinder and Cylinder

The line of intersection between cylinders will be curved since the lateral surfaces of cylinders are curved. Points on the intersection line can be located by either line method or cutting plane method. For plotting an accurate curve, the points at which the curve changes direction must also be located. These points lie on the outermost or extreme lines of each cylinder pierce the surface of the other cylinder. For obtaining the intersection of cylinder and cylinder, cutting plane method is more useful. In this technique, a series of horizontal cutting planes are assumed to be passing through the lines of the horizontal cylinder and these planes cut both cylinders as shown in figure 3. After each sectioning, the top view of the section of the horizontal cylinder will be a rectangle with its width depending on the position of the cutting plane, where as the top view of the vertical cylinder will be a circle with diameter equal the diameter of the vertical cylinder. The points at which the sides of the rectangles intersect the circle will lie on the curve of intersection. The procedure is illustrated through problem 2.

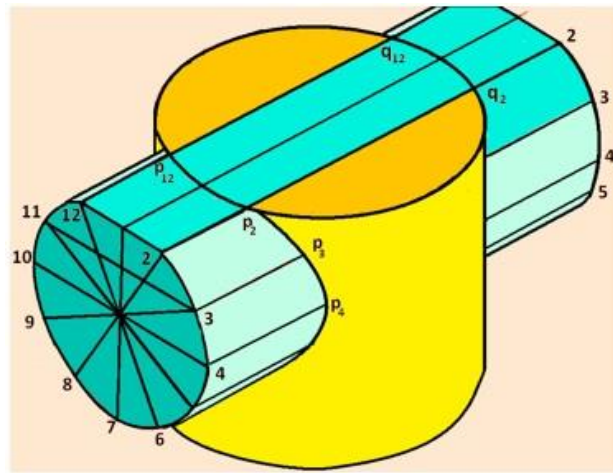


Figure 3 illustrating the cutting plane method for obtaining the curve of intersection.

Problem 2

A vertical cylinder of 80 mm diameter is completely penetrated by another cylinder of 60 mm diameter, their axes bisecting each other at right angles. Draw their projections showing curves of penetration, assuming the axis of the penetrating cylinder to be parallel to the VP.

Solution:

The solution is shown in figure 3.

- The Front view, top view and side view of the two cylinders are drawn without the intersection lines.
- Divide the circumference of the circle in the side view in to 12 equal parts.
- Draw horizontal projectors from these points on to the front view.
- Project the same points from the side view on to the top view and obtain lines 1-1, 2-2, 3-3 , etc. Let us now consider a horizontal section plane passing through points 2-2 and 12-12 (shown in figure 3).
- In the front view, it will be seen as a line coinciding with line 2' 2'.
- In the top view, the section of the horizontal cylinder will be a rectangle of width (i.e. the line 2-12).
- The section of the vertical cylinder will be a circle.

- Points p_2 and p_{12} at which the sides (2-2 and 12-12) of the rectangle cuts the circle will lie on the curve of intersection.
- This point is first obtained in the top view by the intersecting point of line 12-12 and 2-2 with the circle.
- Vertical projector lines are drawn from these points to the front view so as to intersect with the horizontal projectors drawn through points 2 and 12 in the side view to obtain P_2' and P_{12}' in the front view.
- Other cutting planes are also assumed passing through 3-11, 4-10, etc and the procedure repeated to obtain other points p_3' , p_4' , p_5' , etc.
- Similar procedure is adopted to obtain points q_1' , q_2' , etc on the right hand side. S
- Since the axis to the two cylinders intersect, points p_2' and p_{12}' will coincide and hence cannot be shown in the figure.

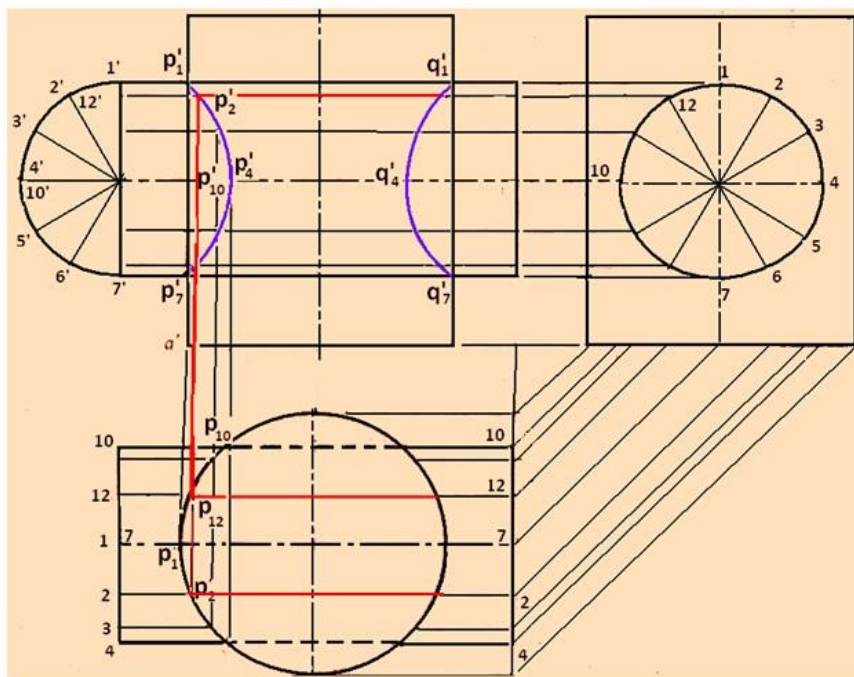


Figure 4. Solution to problem 2.

Intersection of Cone and Cylinder

Figure 5 illustrate the cutting plane method for obtaining the curve of intersection. In this the top view of the cone after sectioning will be a circle and that of the cylinder will be a rectangle with width W . The diameter of the circle and the width of the rectangle in the top view will depend on the height of the section from the cone base. At each height, the intersection points P_2 , P_3 , P_4 ... are obtained and finally these points are joined by a smooth curve in the front view.

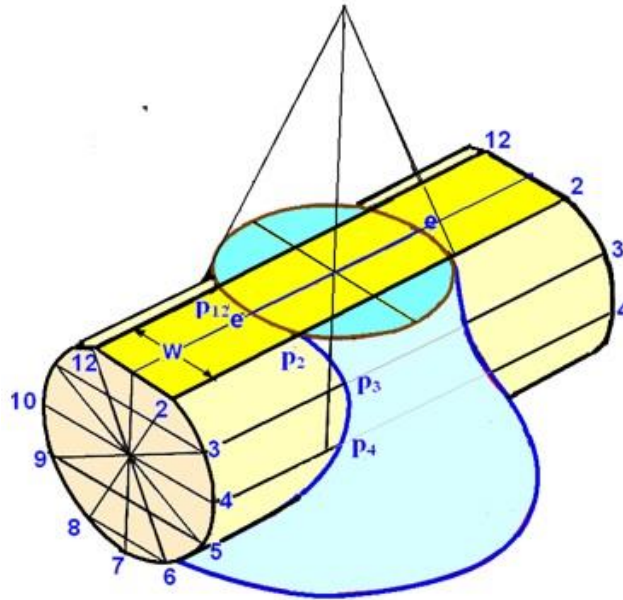


Figure 5 illustrating the cutting plane method for obtaining the curve of intersection.

Problem 3

Example - A vertical cone, diameter of base 75 mm and axis 100 mm long, is completely penetrated by a cylinder of 45 mm diameter. The axis of the cylinder is parallel to HP and VP and intersects the axis of the cone at a point 22 mm above the base. Draw the projections of the solids showing curves of intersection.

Solution

- Draw lines dividing the surface of the cylinder into twelve equal parts.
- Assume a horizontal cutting plane passing through say, point 2.
- The section of the cylinder will be a rectangle of width w (i.e. the line 2-12), while that of the cone will be a circle of diameter ee . These two sections intersect at points p_2 and p_{12} .
- These sections are clearly indicated in the top view by the rectangle 2-2-12-12 and the circle of diameter ee . In the front view, the cutting plane is seen as a line coinciding with $2'2'$.
- Points p_2 and p_{12} when projected on the line $2'2'$ (with which the line $12'-12'$ coincides) will give a point p_2' (with which p_{12}' will coincide).
- Then p_2' and p_{12}' are the points on the curve of intersection.
- To obtain the points systematically, draw circles with centre O and diameters dd, ee, ff , etc. cutting lines through 1, 2 and 12, 3 and 11 etc. at points p_1, p_2 and p_{12}, p_3 and p_{11} etc.
- Project these points to the corresponding lines in the front view.

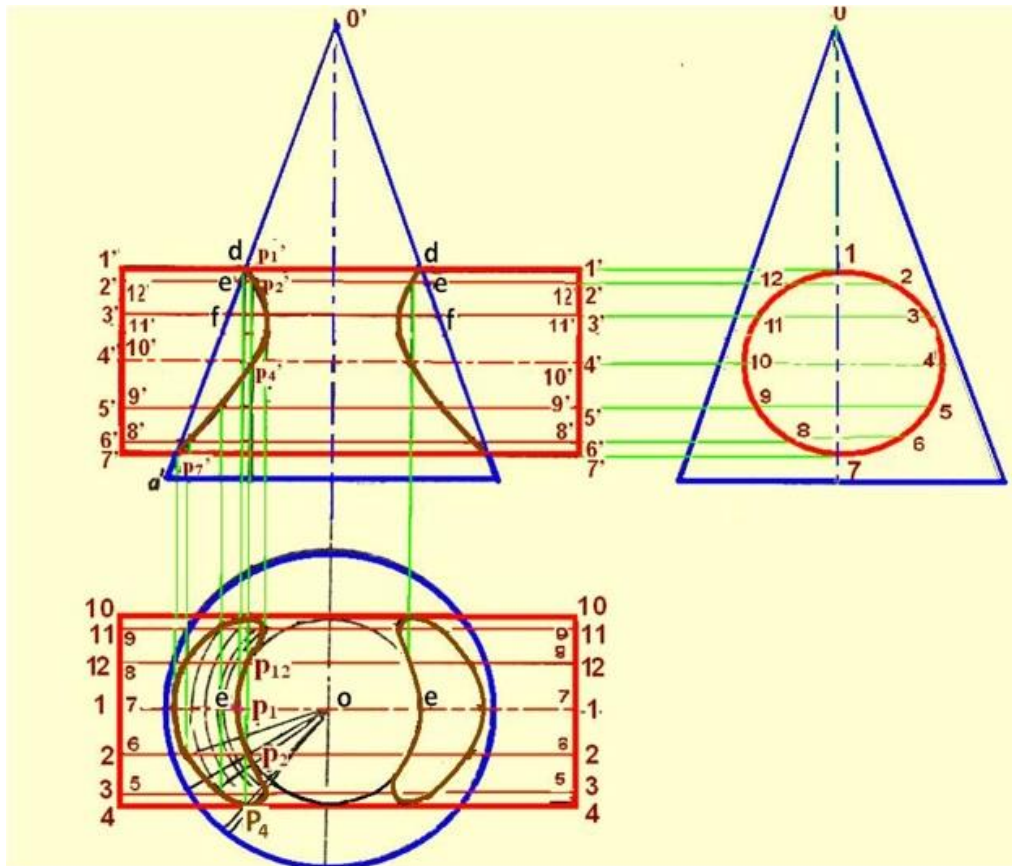


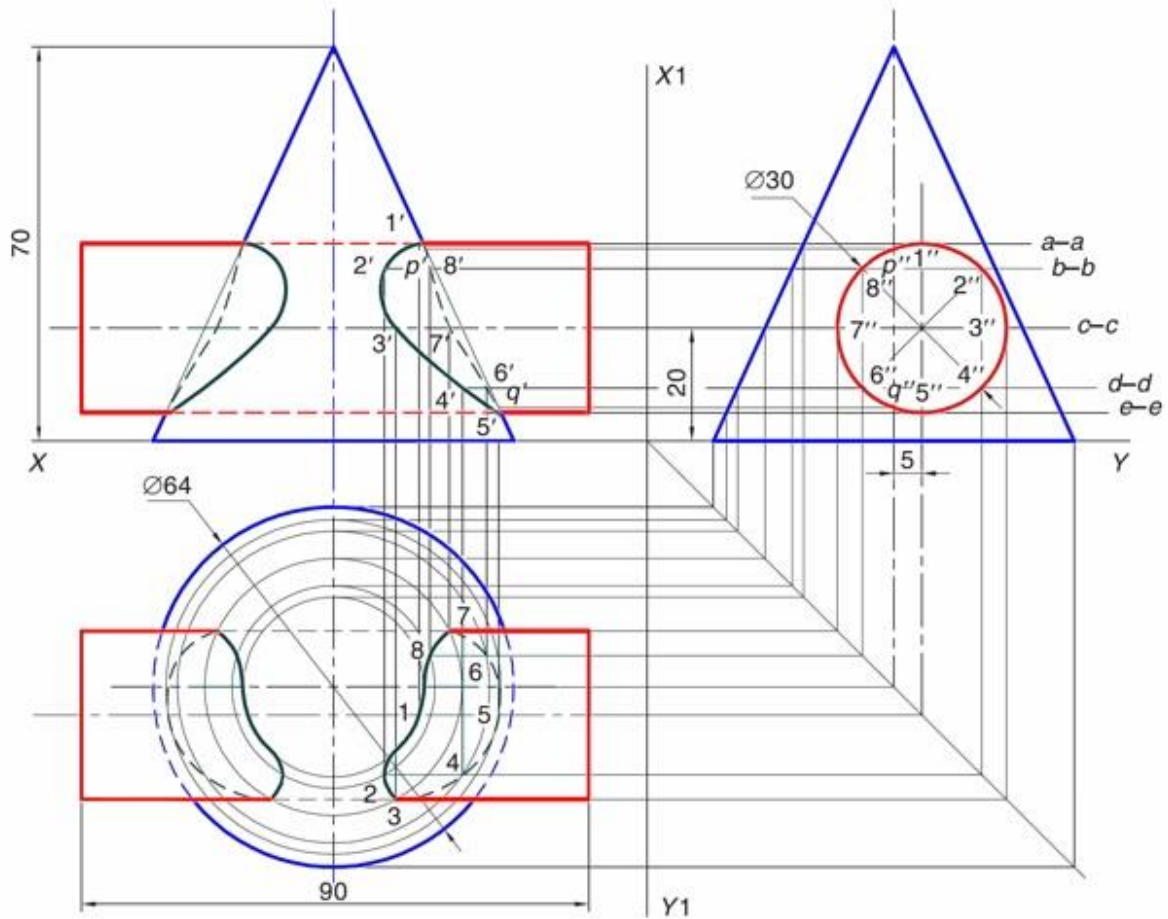
Figure 6. solution to problem 3.

Intersection of Prism and Cylinder

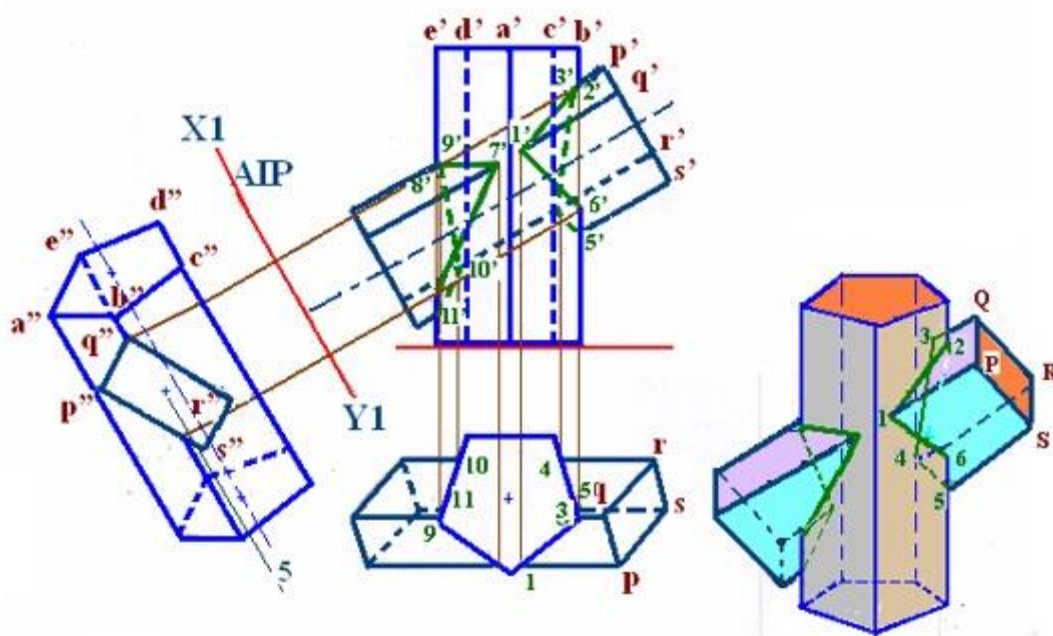
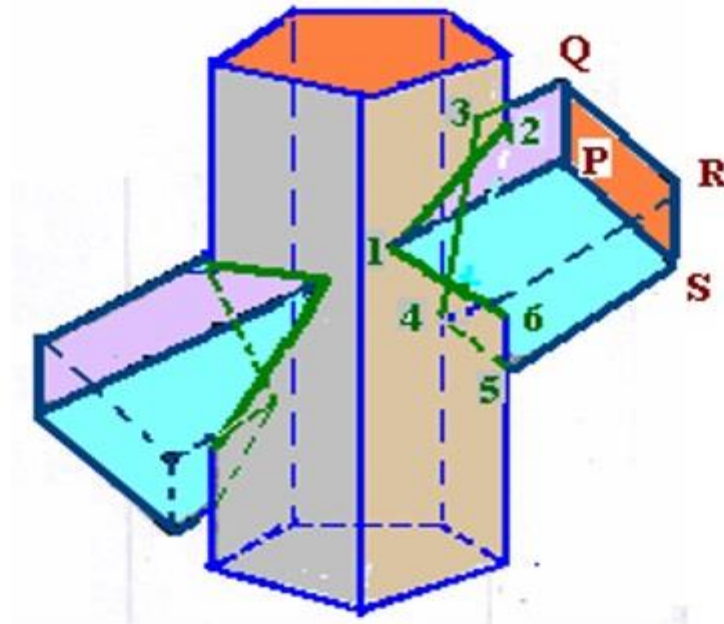
A vertical cylinder with a 60 mm diameter is penetrated by a horizontal square prism with a 40 mm base side, the axis of which is parallel to the VP and 10 mm away from the axis of the cylinder. A face of the prism makes an angle of 30° with the HP. Draw their projections showing curves of intersection.

Intersection of Cylinder and Cone

A cone with a base diameter of 64 mm and an axis length of 70 mm is kept on its base on the HP. A cylinder of diameter 30 mm and length 90 mm penetrates the cone horizontally. The axis of the cylinder is 20 mm above the base of the cone and 5 mm away from the axis of the latter. Draw the three views of the solids showing curve of intersection.

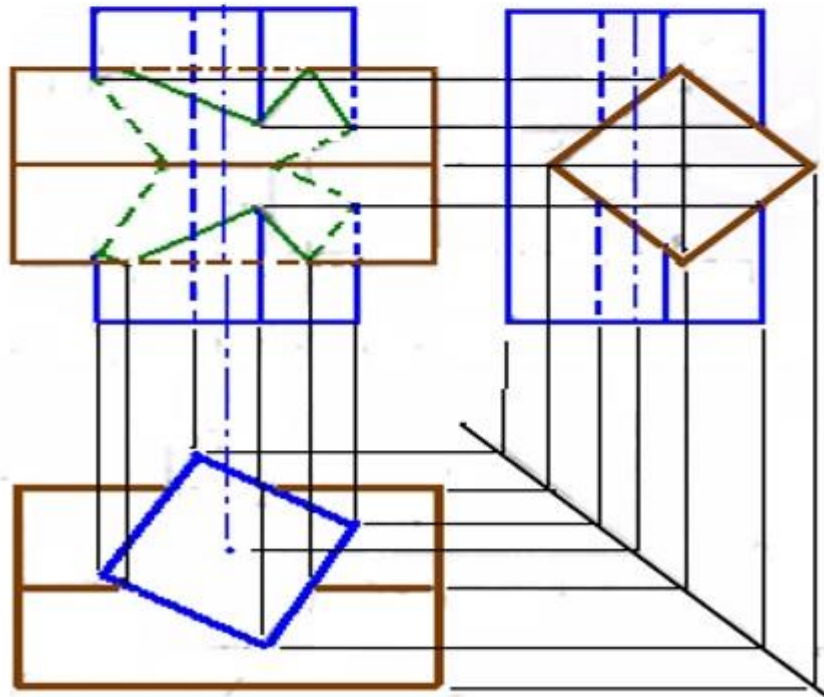
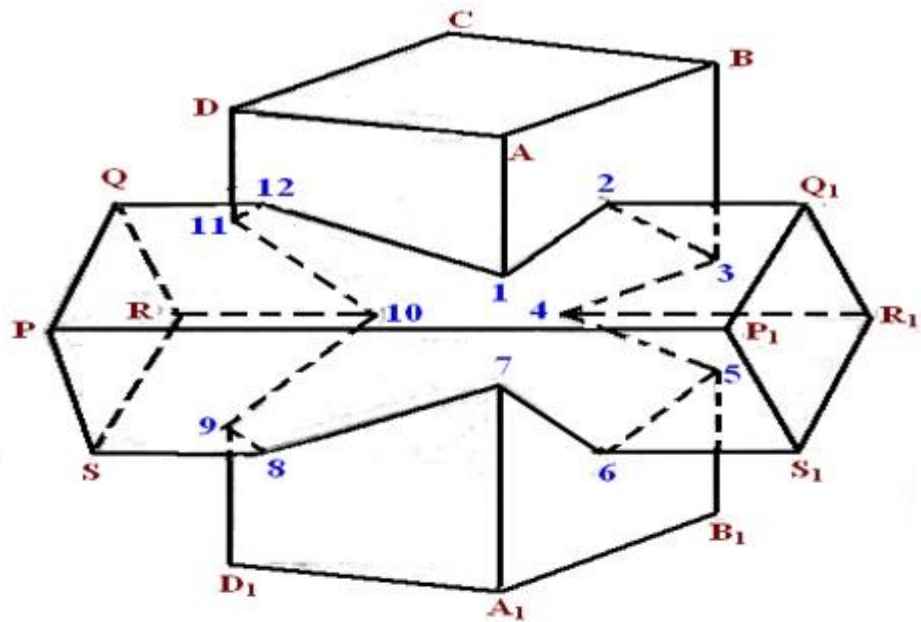


A vertical pentagonal prism 30 mm edge of base and height 100 mm has one of its rectangular faces parallel to VP and nearer to it. It is penetrated by a rectangular prism of side 40mm x 20 mm and 100 mm high, with its front largest lower front rectangular face inclined at 60° to HP. The axis of the rectangular prism is inclined at 30° to HP and parallel to VP, 5 mm in front of the axis of the pentagonal prism and appears to bisect it in the front view. Draw the interpenetration line.



Problem:

A square prism of 40 mm edge of base and 90 mm high rests vertically with its base on HP such that the front right vertical rectangular face is inclined at 60° to VP. This prism is penetrated by another horizontal square prism whose rectangular faces make equal inclination with both HP and VP. The axis of the horizontal prism is passing at the mid height at a distance of 10 mm in front of the vertical prism. The horizontal square prism is of the same dimensions as that of the vertical square prism. Draw the lines of intersection.



A vertical cylinder of 40 mm diameter and 80 mm high is intersected by another cylinder of 35 mm diameter and 80 mm long. The axis of the penetrating cylinder is inclined at 30° to HP, parallel to VP, 6 mm in front of the vertical cylinder and appears to bisect it in front view. Draw the intersection curve.

